R Homework 3: FRED Activity

ADAN/ADEC 7910 Software Tools

Jason F. Sauer

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Background

The St. Louis branch of the Federal Reserve System maintains an extensive database of U.S. economic data that is available to the public. Economists, businesses, and financial institutions rely on the Federal Reserve Economic Data (FRED) database to guide their organizations' behaviors.

Objective

In this activity we use the fredr package to pull FRED data directly into R, then use tidyverse and ggplot2 to analyze and visualize it. The goal is to understand how the federal funds rate (FFR), long-term Treasury yields, inflation, unemployment, and the business cycle relate to one another and to the primary mortgage market.

Connecting to FRED in RStudio

The fredr, tidyverse, and lubridate packages are loaded and the FRED API key is set in chunks hidden from this report (`include = FALSE`), so the key is not published.

Analysis window

Every plot covers the same rolling 20-year window, ending at the date the document is knit.

```
end_date   <- Sys.Date()
start_date <- floor_date(end_date %m-% years(20), unit = "month")

start_date
```

```
## [1] "2006-10-01"
```

```
end_date
```

```
## [1] "2026-10-24"
```

Helper functions

Two user-defined functions keep the code for tasks 1-4 short and consistent. `get_series()` downloads one FRED series and tags it with a readable label; `plot_series()` draws a formatted line chart.

```
get_series <- function(series_id, label, ...) {
  fredr( series_id = series_id, observation_start = start_date, observation_end = end_date,
    ...) %>%
    select(date, value) %>%
    filter(!is.na(value)) %>%
    mutate(series = label)
}

plot_series <- function(df, title, y_label, colors) {
  ggplot(df, aes(x = date, y = value, color = series)) + geom_line(linewidth = 0.8) +
    scale_color_manual(values = colors) + scale_x_date(date_breaks = "2 years", date_labels = "%Y") + la
}
```

Search for Data

```
popular_funds_series <- fredr_series_search_text( search_text = "federal funds", order_by    = "popular

string_of_results <- paste(popular_funds_series$id, collapse = ", ")
print(paste("The top series id are:", string_of_results))
```

```
## [1] "The top series id are: FEDFUNDS, RRPONTSYD, DFF, DFEDTARU, EFFR"
```

The series used in this report are:

Task	Measure	FRED Series ID	Frequency
1	Effective federal funds rate	FEDFUNDS	Monthly
2	10-year Treasury constant maturity	GS10	Monthly
3	CPI, all items (headline)	CPIAUCSL	Monthly, % change from year ago
3	CPI less food and energy (core)	CPILFESL	Monthly, % change from year ago
4	Civilian unemployment rate	UNRATE	Monthly
5	NBER recession indicator	USREC	Monthly (1 = recession)
6	30-year fixed mortgage average	MORTGAGE30US	Weekly, averaged to monthly

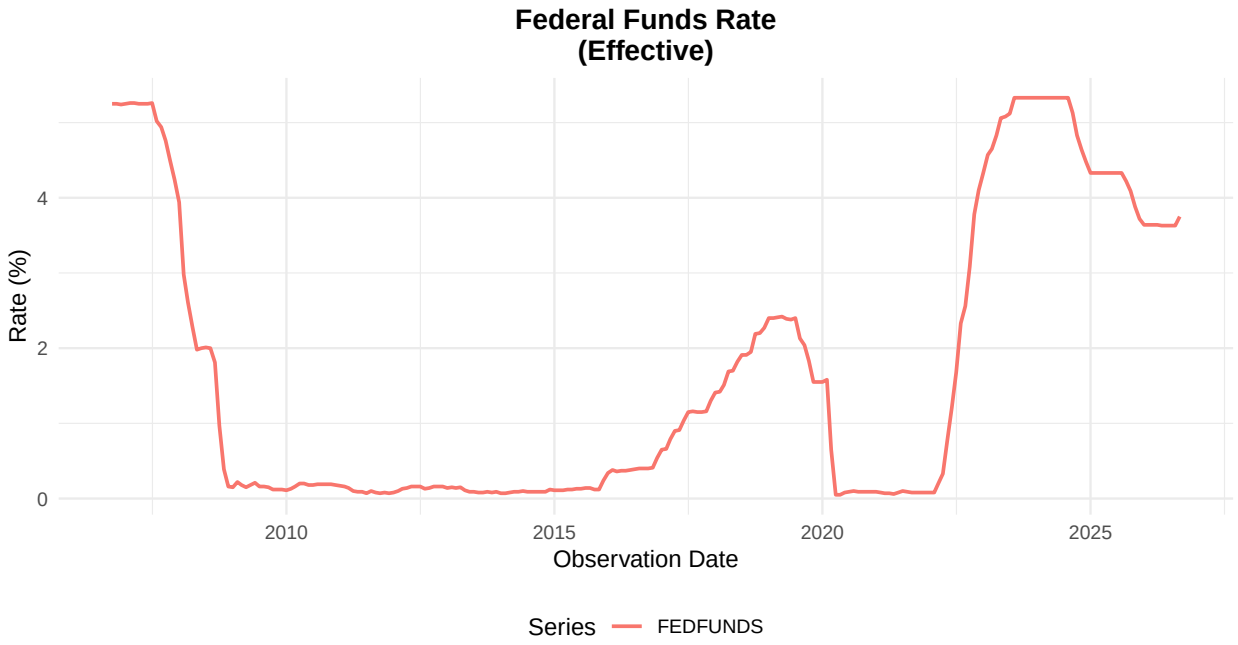
Analysis Tasks

You are a financial analyst at a bank that sells mortgages on the primary market. Your manager wants to understand how historical data can serve as an indicator of the federal funds rate (FFR). The FFR is one of several factors that drive the prime rate for home mortgages. Tasks 1-6 below plot the FFR, the 10-year Treasury rate, inflation, unemployment, and U.S. recessions over the past 20 years, then combine them into a single money-plot used to analyze the mortgage market and forecast the FFR.

Task 1: Federal Funds Rate, Past 20 Years

```r
# Save the #1 search result's series ID, as in the template ("FEDFUNDS")
popular_funds_series_id <- popular_funds_series$id[1]

# Template pipe example, with dates updated to the past 20 years
popular_funds_series_id %>%
  fredr( observation_start = start_date, observation_end = end_date) %>%
  ggplot(data = ., mapping = aes(x = date, y = value, color = series_id)) + geom_line(linewidth = 0.8) +
caption = "Source: FRED, Federal Reserve Bank of St. Louis") + ggtitle("Federal Funds Rate\n(Effective)
```




```r
# Save the same series for the money-plot in Task 6
FFR_data <- get_series(popular_funds_series_id, "Effective Fed Funds Rate")
```

The FFR shows three distinct regimes: a pre-crisis peak above 5% in 2006-07, more than a decade pinned near zero after the 2008 financial crisis (with a brief, modest hiking cycle in 2016-18), and the aggressive 2022-23 tightening cycle that took the rate back above 5% before cuts resumed in late 2024 and 2025.

Task 2: 10-Year Treasury Rate, Past 20 Years

```r
T10_data <- get_series("GS10", "10-Year Treasury")

plot_series(T10_data, title = "10-Year Treasury Constant Maturity Rate", y_label = "Yield (%)", colors
```



Source: FRED, Federal Reserve Bank of St. Louis

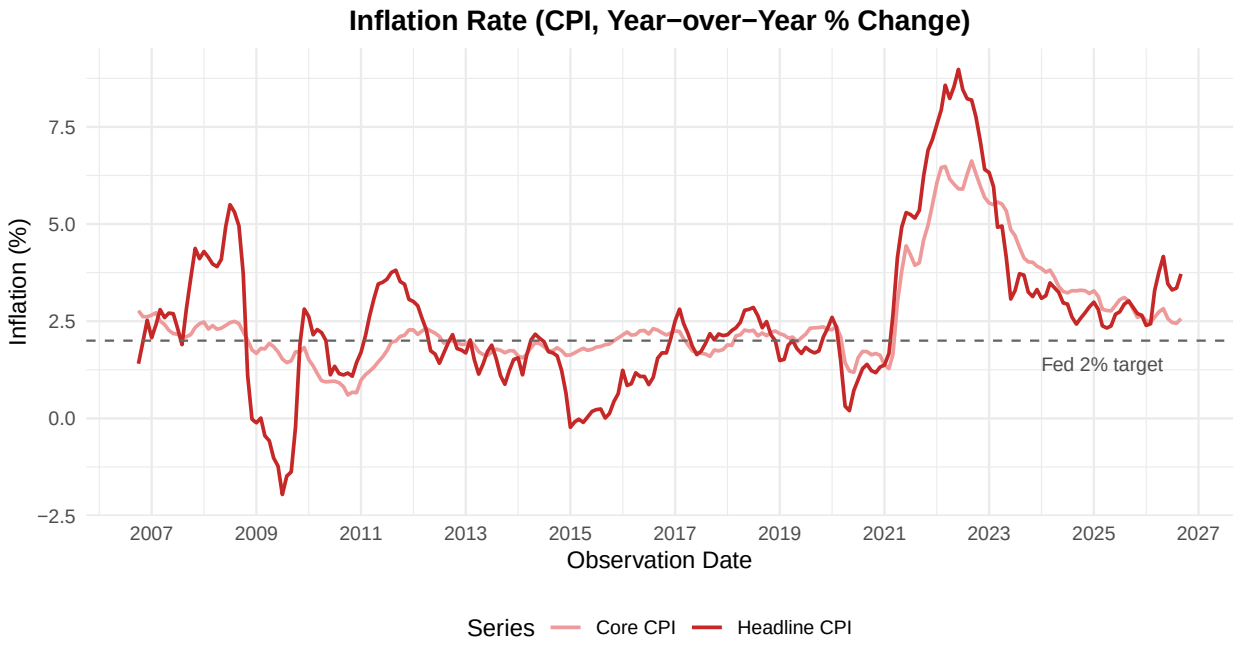
The 10-year yield is set by the bond market rather than the Fed. It reflects expected future short rates, expected inflation, and a term premium. It moved far less than the FFR during the zero-rate years and reached its low in 2020 before rising sharply alongside inflation.

Task 3: Inflation Rate, Past 20 Years

Inflation is measured as the percent change in the Consumer Price Index from one year earlier. Core CPI removes volatile food and energy prices and is a better gauge of underlying trend.

```
CPI_data <- bind_rows( get_series("CPIAUCSL", "Headline CPI", units = "pc1"), get_series("CPILFESL", "C

plot_series(CPI_data, title = "Inflation Rate (CPI, Year-over-Year % Change)", y_label = "Inflation (%)
```

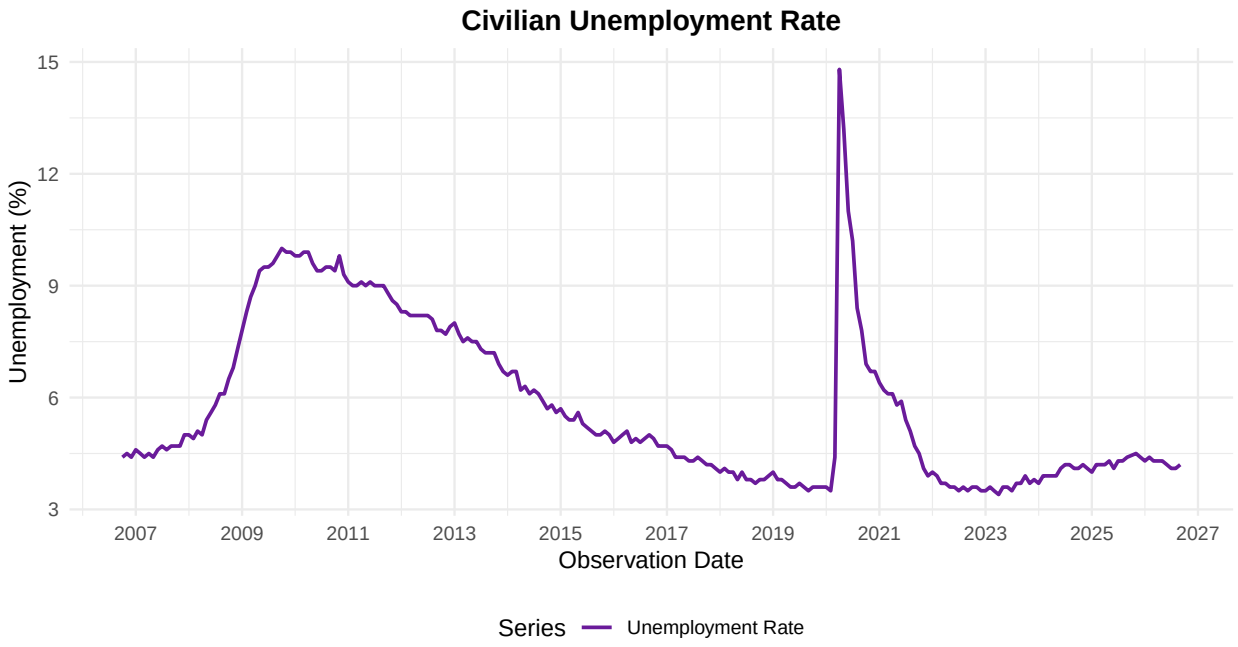


Headline inflation briefly turned negative in 2009 as energy prices collapsed, ran close to the 2% target for most of the 2010s, then surged to roughly 9% in mid-2022, the highest in four decades. It has since come down but has not returned to target.

Task 4: Unemployment Rate, Past 20 Years

```
UNRATE_data <- get_series("UNRATE", "Unemployment Rate")

plot_series(UNRATE_data, title = "Civilian Unemployment Rate", y_label = "Unemployment (%)",  colors  =
```



Unemployment peaked near 10% after the Great Recession and took most of a decade to recover. The 2020 pandemic spike to about 15% was the sharpest in the post-war record but also the fastest recovery.

Task 5: Timeline of U.S. Recessions, Past 20 Years

FRED's `USREC` series equals 1 in months the National Bureau of Economic Research (NBER) classifies as recessionary. The code below turns that monthly 0/1 series into start and end dates for each recession.

```
USREC_data <- get_series("USREC", "Recession")

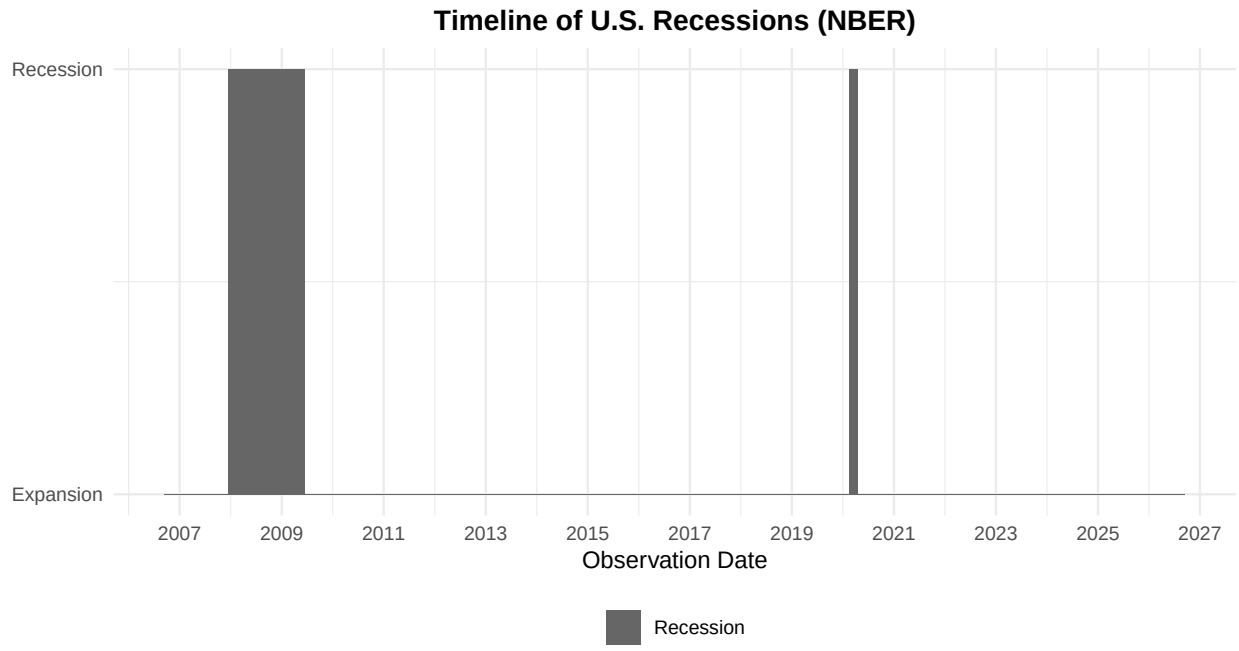
# Number each run of 0s and 1s, keep the 1s, and find each recession's dates
recessions <- USREC_data %>%
  mutate(spell = cumsum(value != lag(value, default = 0))) %>%
  filter(value == 1) %>%
  group_by(spell) %>%
  summarise(start = min(date), end = max(date) %m+% months(1))

knitr::kable(
  recessions %>%
    transmute(`Recession start` = format(start, "%b %Y"), `Recession end` = format(end - 1, "%b %Y")), c
```

Table 1: NBER-dated U.S. recessions in the analysis window

Recession start	Recession end
Jan 2008	Jun 2009
Mar 2020	Apr 2020

```
ggplot(USREC_data, aes(x = date, y = value, fill = series)) + geom_col(width = 31) + scale_fill_manual(
```



The window contains two recessions: the Great Recession (December 2007 peak to June 2009 trough, about 18 months) and the very short COVID-19 recession (February 2020 peak to April 2020 trough). In both cases the Fed cut the FFR to near zero, and in both cases unemployment rose sharply.

Task 6: The Money-Plot

Additional data: the 30-year mortgage rate

Because the manager's question is about the mortgage market, the money-plot adds Freddie Mac's 30-year fixed mortgage average. FRED publishes it weekly, so we ask FRED to average it to monthly to match the other series.

```
MORT_data <- get_series("MORTGAGE30US", "30-Year Mortgage", frequency = "m", aggregation_method = "avg"
```

Building the money-plot

The money-plot is built in labeled steps. Each step adds one layer to the plot object, and the final line displays the finished chart.

```
# Combine the five series into one data set for plotting
money_df <- bind_rows(
  FFR_data    %>% mutate(series = "Fed Funds Rate"),
  T10_data    %>% mutate(series = "10-Year Treasury"),
  MORT_data   %>% mutate(series = "30-Year Mortgage"),
  CPI_data    %>% filter(series == "Headline CPI") %>%
                  mutate(series = "CPI Inflation (YoY)"),
```

```r
  UNRATE_data %>% mutate(series = "Unemployment Rate")) %>%
  mutate(series = factor(series, levels = c("30-Year Mortgage", "10-Year Treasury", "Fed Funds Rate", "

# Color and line style for each series
money_colors <- c("30-Year Mortgage" = "#e65100", "10-Year Treasury" = "#2e7d32", "Fed Funds Rate" = "#

money_lines <- c("30-Year Mortgage" = "solid", "10-Year Treasury" = "solid", "Fed Funds Rate" = "solid"

# Step A: Pipe the data into ggplot and map the columns
money_plot <- money_df %>%
  ggplot(aes(x = date, y = value, color = series, linetype = series))

# Step B: Shade the recessions in the background
money_plot <- money_plot + geom_rect(data = recessions, inherit.aes = FALSE, aes(xmin = start, xmax = en

# Step C: Add a zero reference line and the five data lines
money_plot <- money_plot + geom_hline(yintercept = 0, color = "grey60", linewidth = 0.3) +  geom_line(li

# Step D: Set the colors, line styles, and recession shading color
money_plot <- money_plot + scale_color_manual(values = money_colors) + scale_linetype_manual(values = mo

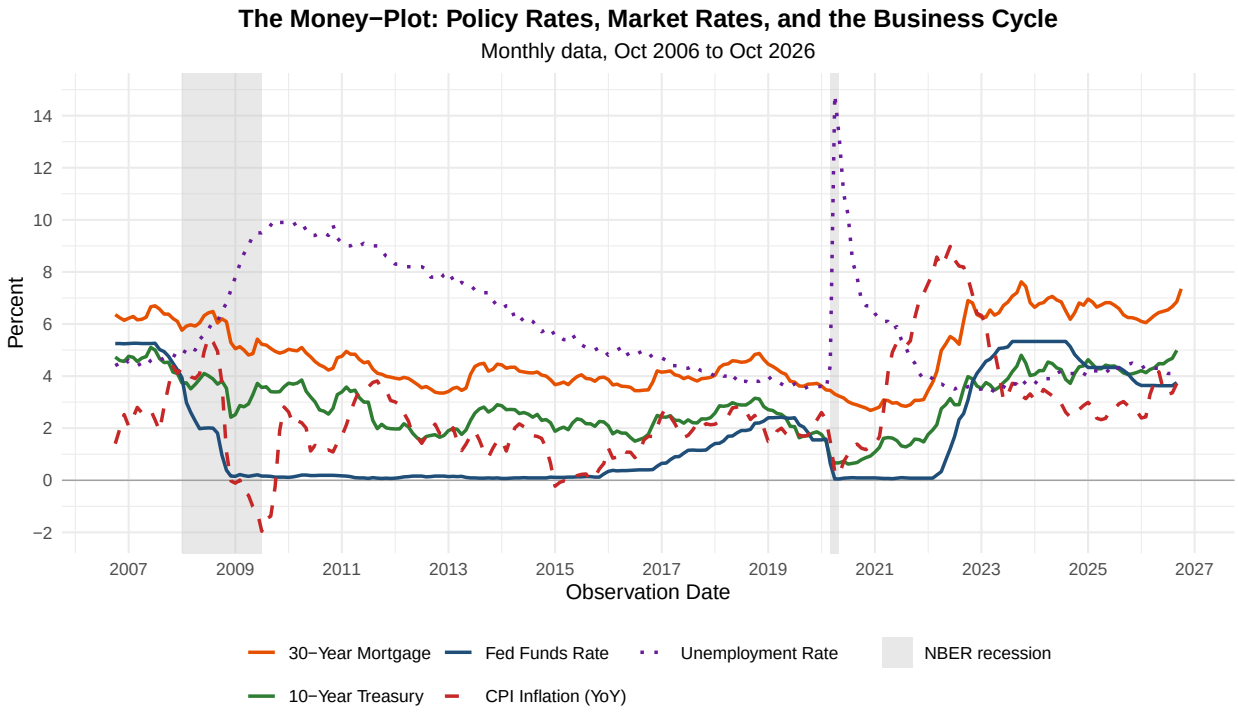
# Step E: Format the date and percent axes
money_plot <- money_plot + scale_x_date(date_breaks = "2 years", date_labels = "%Y") + scale_y_continuou

# Step F: Add the title, subtitle, axis labels, and source caption
plot_subtitle <- paste("Monthly data,", format(start_date, "%b %Y"), "to", format(max(money_df$date), "

money_plot <- money_plot + labs(title    = "The Money-Plot: Policy Rates, Market Rates, and the Business
       subtitle = plot_subtitle,
       x        = "Observation Date",
       y        = "Percent",
       color    = NULL,
       linetype = NULL,
       fill     = NULL,
       caption  = "Sources: FRED (FEDFUNDS, GS10, MORTGAGE30US, CPIAUCSL, UNRATE, USREC)")

# Step G: Arrange the legend and apply the theme
money_plot <- money_plot + guides(color = guide_legend(nrow = 2), linetype = guide_legend(nrow = 2)) +

# Step H: Display the finished plot
money_plot
```



Supporting numbers

```
# Put each series in its own column, keeping months where all five exist
wide <- money_df %>%
  select(date, series, value) %>%
  pivot_wider(names_from = series, values_from = value) %>%
  drop_na()

cor_mort_10y  <- cor(wide$`30-Year Mortgage`, wide$`10-Year Treasury`)
cor_mort_ffr  <- cor(wide$`30-Year Mortgage`, wide$`Fed Funds Rate`)
spread        <- wide$`30-Year Mortgage` - wide$`10-Year Treasury`
avg_spread    <- mean(spread)
latest_spread <- last(spread)

knitr::kable(tibble(Measure = c("Correlation: mortgage rate vs. 10-year Treasury",  "Correlation: mortg
```

Table 2: How closely mortgage rates track each benchmark

Measure	Value
Correlation: mortgage rate vs. 10-year Treasury	0.95
Correlation: mortgage rate vs. fed funds rate	0.84
Average mortgage minus 10-year spread (pct. points)	1.92
Latest mortgage minus 10-year spread (pct. points)	1.87

```r
money_df %>%
  group_by(series) %>%
  slice_max(date, n = 1) %>%
  ungroup() %>%
  transmute(Series = series, `Latest month` = format(date, "%b %Y"), `Value (%)` = value) %>%
  knitr::kable(digits = 2, caption = "Most recent observation for each series")
```

Table 3: Most recent observation for each series

Series	Latest month	Value (%)
30-Year Mortgage	Oct 2026	7.35
10-Year Treasury	Sep 2026	4.99
Fed Funds Rate	Sep 2026	3.75
CPI Inflation (YoY)	Sep 2026	3.72
Unemployment Rate	Sep 2026	4.20

Why the money-plot is built this way

The plot puts every series on one shared percent axis so that the reader can compare levels directly: how far the mortgage rate sits above the 10-year Treasury, whether the policy rate is above or below inflation, and where unemployment is relative to its history. The 30-year mortgage rate is added because it is the price our bank actually sells, so the question "what drives it?" should be answerable from the picture. Recessions are shaded in the background rather than drawn as a line, so they frame the story without competing with the rates. Market and policy rates are drawn as solid lines, while the two macro indicators the Fed reacts to (inflation and unemployment) are dashed and dotted, which separates "what the Fed watches" from "what borrowers pay." Unemployment and inflation are kept on the same axis instead of a second axis, because dual axes make it easy to imply relationships that are only an artifact of scaling.

Analysis of the mortgage market

The Fed sets the short end; the bond market sets mortgage rates. The 30-year mortgage rate moves almost in lockstep with the 10-year Treasury (correlation of 0.95), and much less tightly with the FFR (0.84). Mortgages are priced off long-term yields because most 30-year loans are refinanced or paid off in roughly a decade and are funded through mortgage-backed securities that compete with Treasuries for investors. Over the window the mortgage rate averaged about 1.92 percentage points above the 10-year yield; the latest spread is 1.87 points.

The FFR still matters, through expectations. The FFR influences the 10-year yield through expected future short rates, and the plot shows this clearly in the 2022-23 cycle: as the Fed raised the FFR from near zero to above 5%, the 10-year and the mortgage rate roughly doubled to their highest levels in two decades. The reverse case is just as instructive. The Fed cut the FFR by 1.75 percentage points between September 2024 and December 2025, yet mortgage rates barely fell, because long-term yields stayed high on persistent inflation and heavy Treasury issuance. Cutting the FFR does not guarantee cheaper mortgages.

Recessions are when the FFR falls fastest. In both shaded periods the Fed cut aggressively as unemployment rose, and mortgage rates eventually followed long yields down, which set off refinancing booms in 2009-12 and 2020-21. Rising unemployment is the most reliable historical signal that the FFR is about to fall.

Real rates explain the Fed's stance. When CPI inflation is above the FFR (most of 2021-22), policy is loose; when the FFR is above inflation (2023-24), policy is restrictive. Today the two lines sit close together, which means policy is only mildly restrictive in real terms, and the Fed has responded to the recent re-acceleration in inflation by leaning toward tighter policy rather than looser.

Where things stand (October 2026)

The latest data paint a stagflation-tinged picture. Headline CPI inflation rose to 3.7% year-over-year in September 2026, up from 3.4% the month before and well above the 2% target. At the same time, the labor market is cooling: payrolls grew by only about 29,000 in September and unemployment rose to 4.2%. Despite the soft jobs data, the FOMC raised the target range by 25 basis points to 3.75%-4.00% on September 16, 2026, its first hike since July 2023, and the accompanying projections signaled roughly one more increase before year-end. In the bond market, the 10-year yield climbed above 5%, its highest level since 2007, and Freddie Mac's 30-year mortgage average reached 7.40% in mid-October, up from about 6.3% a year earlier.

Prediction: the FFR and the mortgage market

It is not possible to know the future, but the plotted history supports a reasoned base case.

Federal funds rate. The Fed is fighting rising inflation while the labor market softens, a conflict between its two mandates. Historically the Fed has prioritized inflation once it runs well above target (as in 2022-23), so the base case is one more 25 basis point hike by December 2026, taking the target range to 4.00%-4.25%, followed by a hold through at least the first half of 2027. Rate cuts become likely only if either inflation clearly turns down or unemployment rises meaningfully further. The money-plot shows that once unemployment starts climbing quickly, the Fed has historically reversed course within months, so a rise toward 5% unemployment is the main risk to this forecast and would pull cuts forward into 2027.

Mortgage rates. Because mortgage rates track the 10-year Treasury, the near-term outlook depends on long yields, not the Fed's next move. With inflation re-accelerating and the 10-year above 5%, the best estimate is that the 30-year rate stays in roughly a 6.75%-7.75% range over the next six to twelve months. A meaningful decline below 6.5% would most likely require a growth scare or recession that pushes investors into Treasuries.

Implications for the bank. Expect weak refinance volume, since most existing borrowers hold rates well below today's market and have no incentive to refinance (the "lock-in" effect). Purchase volume should also stay subdued, because high rates on top of high home prices strain affordability. Adjustable-rate and buydown products are likely to gain share as borrowers look for lower initial payments. The bank should be positioned to capture a refinancing wave quickly if a recession pushes long rates down, as happened after each shaded period on the money-plot, while managing rate-lock pipeline risk carefully in the meantime given how volatile the 10-year has been.